CA II Equation Sheet

1 Power Screws

#	Equation
1.1	$d_r = d - p$
1.2	$d_m = d - p/2$
1.3	$\ell = np$
1.4	$w = p/2$
1.5	$t = p/2$
1.6	$\alpha_{sq} = 0°$
1.7	$\alpha_{ac} = 14.5°$
1.8	$T_R = \frac{Fd_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$
1.9	$T_L = \frac{Fd_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$
1.10	$T_C = F f_c d_c/2$
1.11	$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$

2 Compression Springs

#	Equation
2.1	$F = kx$
2.2	$C = D/d$
2.3	$K_B = \frac{4C+2}{4C-3}$
2.4	$\tau_m = K_B \frac{8FD}{\pi d^3}$
2.5	$K_F = 0.45$
2.6	$\tau_y = K_F \frac{A}{d^m}$
2.7	$k = \frac{d^4 G}{8D^3 N_a}$
2.8	(1) $4 \le C \le 12$
2.9	(2) $3 \le N_a \le 15$
2.10	(3) $L_0 \le 5.26D$
2.11	(4) $x \le \frac{7}{8}(L_0 - L_s)$
2.12	(5) $n_s = 1.2 \le \frac{\tau_y}{\tau_m}$

Table 2.1: Spring end type parameters

Parameter	Plain	Ground	Squared	Squared & Ground
N_a	N_t	$N_t - 1$	$N_t - 2$	$N_t - 2$
p	$(L_0 - d)/N_a$	$L_0/(N_a + 1)$	$(L_0 - 3d)/N_a$	$(L_0 - d)/N_a$
L_s	$d(N_t + 1)$	dN_t	$d(N_t + 1)$	$d(N_t - 1)$

Table 2.2: Spring material properties

Material	A		m	Wire Diameter d		G	
	kpsi·in^m	MPa·mm^m	-	in	mm	Mpsi	GPa
Hard-Drawn wire (A227)	140	1783	0.190	$d < 0.032$	$d < 0.8$	11.7	80.7
				$0.032 \le d < 0.064$	$0.8 \le d < 1.6$	11.6	80.0
				$0.064 \le d < 0.125$	$1.6 \le d < 3.0$	11.5	79.3
				$0.125 \le d$	$3.0 \le d$	11.4	78.6
Music wire (A228)	201	2211	0.145	$d < 0.032$	$d < 0.8$	12.0	82.7
				$0.032 \le d < 0.064$	$0.8 \le d < 1.6$	11.85	81.7
				$0.064 \le d < 0.125$	$1.6 \le d < 3.0$	11.75	81.0
				$0.125 \le d$	$3.0 \le d$	11.6	80.0

3 Gears

#	Equation
3.1	$\eta = \dfrac{T_{out}\omega_{out}}{T_{in}\omega_{in}}$
3.2	$\psi_{spur} = 0^\circ$
3.3	$\cos\psi = \dfrac{\tan\phi_n}{\tan\phi}$
3.4	$p = \dfrac{\pi d}{N}$
3.5	$P = \dfrac{N}{d}$
3.6	$m = \dfrac{N_G}{N_P}$
3.7	$N_P\omega_P = N_G\omega_G$
3.8	$k_{full} = 1$
3.9	$k_{stub} = 0.8$
3.10	$N_P \geq \dfrac{2k\cos\psi\left(m + \sqrt{m^2 + (1+2m)\sin^2\phi}\right)}{(1+2m)\sin^2\phi}$
3.11	$N_G \leq \dfrac{N_P^2\sin^2\phi - 4k^2\cos^2\psi}{4k\cos\psi - 2N_P\sin^2\phi}$
3.12	Rack: $N_P \geq \dfrac{2k\cos\psi}{\sin^2\phi}$

4 Bolts

#	Equation
4.1	(i) $\sum F = 0$ (ii) $\sum M = 0$
4.2	Bearing: $\sigma = \dfrac{F}{A}$
4.3	Shear: $\tau = \dfrac{F}{A}$
4.4	Bending: $\sigma = -\dfrac{My}{I}$
4.5	Torsion: $\tau = \dfrac{Tr}{J}$
4.6	(i) $A_{\bigcirc} = \dfrac{\pi d^2}{4}$ (ii) $A_{\square} = bh$
4.7	(i) $I_{\bigcirc} = \dfrac{\pi d^4}{64}$ (ii) $I_{\square} = \dfrac{bh^3}{12}$
4.8	(i) $J_{\bigcirc} = \dfrac{\pi d^4}{32}$ (ii) $J_{\square} = \dfrac{bh}{12}(b^2+h^2)$
4.9	$\tau_y = \dfrac{\sigma_y}{\sqrt{3}}$
4.10	(i) $n_s = \dfrac{\tau_y}{\tau_m}$ (ii) $n_s = \dfrac{\sigma_y}{\sigma_m}$
4.11	(i) $F_i' = \dfrac{F}{N}$ (ii) $F_i'' = \dfrac{Mr_i}{\sum_i r_i^2}$

Table 4.1: Yield Stress for Standard Steel Bolts and Heat-Treated Steels

Bolt Class	σ_y [MPa]	Bolt Class	σ_y [kpsi]	Steel Class	σ_y [MPa]	[kpsi]
ISO 4.6	240	SAE Grade 1	36	1018 (HR)	220	32
ISO 5.8	420	SAE Grade 4	100	1018 (CD)	370	54
ISO 8.8	660	SAE Grade 7	115	1020 (HR)	210	30
ISO 9.8	720	SAE Grade 8	130	1020 (CD)	390	57

5 Welds

#	Equation	#	Equation	#	Equation	#	Equation
5.1	$\tau^* = \sqrt{\tau^2 + \left(\frac{\sigma_1-\sigma_2}{2}\right)^2}$	5.2	$t = s/\sqrt{2}$	5.3	$\tau_y = 0.4\sigma_y$	5.4	$J = tJ_u, \quad I = tI_u$

5.5	$\vert$ (i) $J_u = \dfrac{h^3}{12}$	(ii) $I_u = \dfrac{h^3}{12}$	(iii) $y^* = \dfrac{h}{2}$
5.6	$=$ (i) $J_u = \dfrac{b(3h^2+b^2)}{6}$	(ii) $I_u = \dfrac{bh^2}{2}$	(iii) $y^* = \dfrac{h}{2}$
5.7	$\sqcap$ (i) $J_u = \dfrac{8h^3+6hb^2+b^3}{12} - \dfrac{h^4}{2h+b}$	(ii) $I_u = \dfrac{h^3(2b+h)}{3(b+2h)}$	(iii) $y^* = \dfrac{h(b+h)}{b+2h}$